OilSense: A Low-Cost Dual-Sensor Instrument for Detecting Frying-Oil Degradation

Abstract

Oil that is reused for deep frying breaks down into polar compounds, and India's food regulator (FSSAI) requires oil above 25 % total polar materials (TPM) to be discarded. Most small food businesses cannot measure TPM. Laboratory tests take days, and portable meters such as the testo 270 cost about ₹55,000, so oil is changed by habit, colour or smell. This project will develop OilSense, a low-cost hand-held instrument that will assess frying-oil degradation at the fryer within seconds. It will measure the oil in two independent ways. An AS7265x 18-channel spectral sensor (410–940 nm) will measure how strongly the oil absorbs light, and an FDC1004 capacitance-to-digital converter with two stainless-steel electrodes will measure its dielectric response. A temperature probe will allow correction, and an ESP32 microcontroller will combine the readings, display the result and upload it to a cloud dashboard. Because the two channels respond to the same degradation through unrelated physical effects, comparing them should reveal when a reading is unreliable, for example when water or food particles are present. The optical signal will be normalised logarithmically, following the Beer–Lambert law, so that it changes linearly with the concentration of degradation products. The probe will be calibrated in air and fresh oil without a reference instrument, then tested on oil heated for increasing times. The project will compare the two channels, measure how probe geometry limits the capacitive channel, and test whether logarithmic normalisation outperforms linear normalisation. The target is a working prototype costing under ₹10,000.

Introduction

Deep frying is common across India, and the same oil is usually reused many times. At 160–190 °C, the oil reacts with water, oxygen and heat to form free fatty acids, oxidised compounds and polymers [1]. Together these are measured as total polar materials (TPM), and FSSAI sets a discard limit of 25 % [2]. Street vendors and small restaurants have no affordable way to check this limit. Low-cost single-sensor devices also have a hidden weakness: they cannot tell a real change in the oil from an error, such as water from food raising a capacitive reading. This project will address both problems with an inexpensive two-channel instrument that checks its own readings.

Innovation

Unlike existing low-cost devices, OilSense will combine optical and capacitive sensing of frying oil and use their agreement as a check on each reading. When the two channels disagree, the direction of the disagreement will suggest the cause: water mainly raises permittivity, while crumbs mainly raise absorbance. The optical signal will also be scaled with the Beer–Lambert law instead of linearly, so that both indicators can be compared fairly.

Methodology

Hardware. An ESP32 will read an AS7265x spectral sensor and an FDC1004 capacitance converter over a shared I²C bus, and a digital temperature probe over 1-Wire. Two threaded grade-304 stainless-steel electrodes, held in a 3D-printed PETG housing with a depth-limiting

collar, will form the sensing capacitor. An OLED will show results, and readings will be uploaded to a Firebase database.

Calibration. The probe will be measured in air ($\varepsilon \approx 1$) and fresh oil ($\varepsilon \approx 3$) to find its cell constant and parasitic capacitance, so no reference meter will be needed.

Samples. Oil will be heated at frying temperature with food added at intervals. Samples will be taken at set times and cooled to a common temperature.

Measurement. Each reading will average 64 capacitance conversions and record all 18 spectral channels. Each sample will be measured at least three times.

Analysis. The optical indicator will be the ratio of a blue channel to a near-infrared channel, normalised both linearly and logarithmically. Both indicators will be scaled from 0 (fresh) to 1 (discard limit) and compared using a tolerance of 0.15. Water and crumbs will be added to test samples to check fault detection.

Expected Results

I expect the optical channel to track degradation clearly, because degraded oil absorbs strongly at blue wavelengths [3]. I expect the capacitive change to be small: a published sensor changed by only 0.24 pF as TPM rose from 7 % to 32.5 % [4], so probe geometry, temperature and immersion depth may limit this channel. Theory predicts that linear normalisation of the optical signal will overstate mid-life degradation by up to 0.14–0.27 of full scale, while logarithmic normalisation should remove this error. The project should show which channel is more reliable, how probe design affects the capacitive signal, and whether a dual-sensor instrument can be built for about one-sixth of the cost of a commercial meter [5].

Acknowledgement and References

I thank my teachers and mentor for their guidance.

[1] E. Choe and D. B. Min, *J. Food Sci.*, vol. 72, no. 5, pp. R77–R86.

[2] FSSAI, Food Safety and Standards (Licensing and Registration) First Amendment Regulations.

[3] X.-Q. Xu, *J. Am. Oil Chem. Soc.*, vol. 77, no. 10, pp. 1083–1086.

[4] M. Liu et al., *Sensors*, vol. 19, no. 24, p. 5375.

[5] Testo SE, testo 270 product specification.